

5. Which of the following statements is ***not*** true for electric energy in ac form compared to that in dc form ?
- (A) Production of ac is economical.
 - (B) ac can be easily and efficiently converted from one voltage to the other.
 - (C) ac can be transmitted economically over long distances.
 - (D) ac is less dangerous.

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12. The rms and the average value of an ac voltage $V = V_0 \sin \omega t$ volt over a cycle respectively will be :

(A) $\frac{V_0}{2}, \frac{V_0}{\sqrt{2}}$

(B) $\frac{V_0}{\pi}, \frac{V_0}{2}$

(C) $\frac{V_0}{\sqrt{2}}, 0$

(D) $V_0, \frac{V_0}{\sqrt{2}}$

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(C) $\frac{V_0}{\sqrt{2}}, 0$

6. An ac voltage is given as $v = 14 \sin(314t)$ V. The average and the effective value of the voltage (in V) over a cycle are respectively :

(A) 14 and 7

(B) 10 and 14

(C) 0 and 10

(D) 10 and 0

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(C) 0 and 10

- (b) The instantaneous value of a voltage (in volt) is given by $E = 140 \sin 314 t$, where t is in seconds. Find :
- (i) rms value of the applied voltage.
 - (ii) the frequency of ac supply.

(ii) (I) $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$	$\frac{1}{2}$
$= \frac{140}{\sqrt{2}} = 100 \text{ volt}$	$\frac{1}{2}$
(II) $\nu = \frac{\omega}{2\pi}$	$\frac{1}{2}$
$= \frac{314}{2(3.14)} = 50 \text{ Hz}$	$\frac{1}{2}$

10. The average value of voltage of 220 V ac mains during its positive half cycle will be :

(A) 156 V

(B) 198 V

(C) 220 V

(D) zero

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(B) 198 V

8. An alternating current is given by $I = I_0 \cos (100\pi)t$. The least time the current takes to decrease from its maximum value to zero will be :

(A) $\left(\frac{1}{200}\right)\text{s}$

(B) $\left(\frac{1}{150}\right)\text{s}$

(C) $\left(\frac{1}{100}\right)\text{s}$

(D) $\left(\frac{1}{50}\right)\text{s}$

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(A) 1/200 s

24. Differentiate between the peak value and root mean square value of an alternating current. Derive the expression for the root mean square value of alternating current, in terms of its peak value.

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3

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Peak value is the maximum value of the alternating current.

$\frac{1}{2}$

rms current is the equivalent dc current that would produce the same average power loss as the alternating current.

$\frac{1}{2}$

Alternatively: -

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

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$$I_{\text{rms}} = 0.707 I_0$$

The instantaneous power dissipated in the resistor is $P = i^2 R = i_m^2 R \sin^2 \omega t$

$\frac{1}{2}$

The average power over a cycle is: -

$$\bar{P} = \langle i^2 R \rangle = \langle i_m^2 R \sin^2 \omega t \rangle$$

$\frac{1}{2}$

$$\langle \sin^2 \omega t \rangle = \frac{1}{2}$$

$$\bar{P} = \frac{1}{2} i_m^2 R = I_{rms}^2 R$$

$\frac{1}{2}$

$$I_{rms} = \sqrt{\frac{i_m^2}{2}} = \frac{i_m}{\sqrt{2}}$$

$\frac{1}{2}$

Alternatively: -

$$i = i_o \sin \omega t$$

$$I_{rms}^2 = \frac{1}{T} \int_0^T i_o^2 \sin^2 \omega t dt$$

$\frac{1}{2}$

$$= \frac{i_o^2}{T} \int_0^T \left(\frac{1 - \cos 2\omega t}{2} \right) dt$$

$\frac{1}{2}$

$$I_{rms}^2 = \frac{i_o^2}{2T} (T - 0) \quad \left[\text{As } \int_0^T \frac{\cos 2\omega t}{2} dt = 0 \right]$$

$\frac{1}{2}$

$$I_{rms} = \frac{i_o}{\sqrt{2}}$$

$\frac{1}{2}$

9. The r.m.s. value of a current given by $i = (i_1 \cos \omega t + i_2 \sin \omega t)$ is –

(A) $\frac{1}{\sqrt{2}} (i_1 + i_2)$

(B) $\frac{1}{\sqrt{2}} (i_1 - i_2)$

(C) $\frac{1}{\sqrt{2}} \sqrt{i_1^2 + i_2^2}$

(D) $\frac{1}{\sqrt{2}} (i_1^2 + i_2^2)$

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(C) $\frac{1}{\sqrt{2}} \sqrt{i_1^2 + i_2^2}$

6. The average value of the alternating voltage $v = (157 \text{ V}) \sin \omega t$ over its first half-cycle is :

(A) 157 V

(B) $\frac{157}{\sqrt{2}} \text{ V}$

(C) 78.5 V

(D) 100 V

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10. A voltage signal is described by :

$$v = V_0 \quad \text{for } 0 \leq t \leq \frac{T}{2}$$

$$= 0 \quad \text{for } \frac{T}{2} \leq t \leq T$$

for a cycle. Its rms value is :

(a) $\frac{V_0}{\sqrt{2}}$

(b) V_0

(c) $\frac{V_0}{2}$

(d) $\sqrt{2} V_0$

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(a) $\frac{V_0}{\sqrt{2}}$
